

Unified Smart Home Architecture Proposal

G+1 villa automation architecture with local-first control, real device stack, schematic diagrams, room-wise automation, and installation-ready topology.



Project Scope

Ground + 1 floor smart home solution for 5 first-floor bedrooms, attached washrooms, living, dining, prayer area, powder room, staircase, exterior zones, and garden.

Design Principle

Local-first automation. Internet improves remote access but must not be required for daily operation.

Document Output

Architecture, schematic diagrams, named product stack, BOQ categories, and implementation phases.

1. Recommended Real Device Stack

Core Automation		
Layer	Recommended Device / Product	Role
Controller	Home Assistant Green or Intel NUC running Home Assistant OS	Main local automation server
Zigbee/Thread	Home Assistant Connect ZBT-2 or SONOFF Zigbee 3.0 USB Dongle Plus ZBDongle-E	Zigbee sensor and device mesh
MQTT	Eclipse Mosquitto	Local command and telemetry broker
Firmware	ESPHome for ESP32-based custom nodes	Custom controller firmware

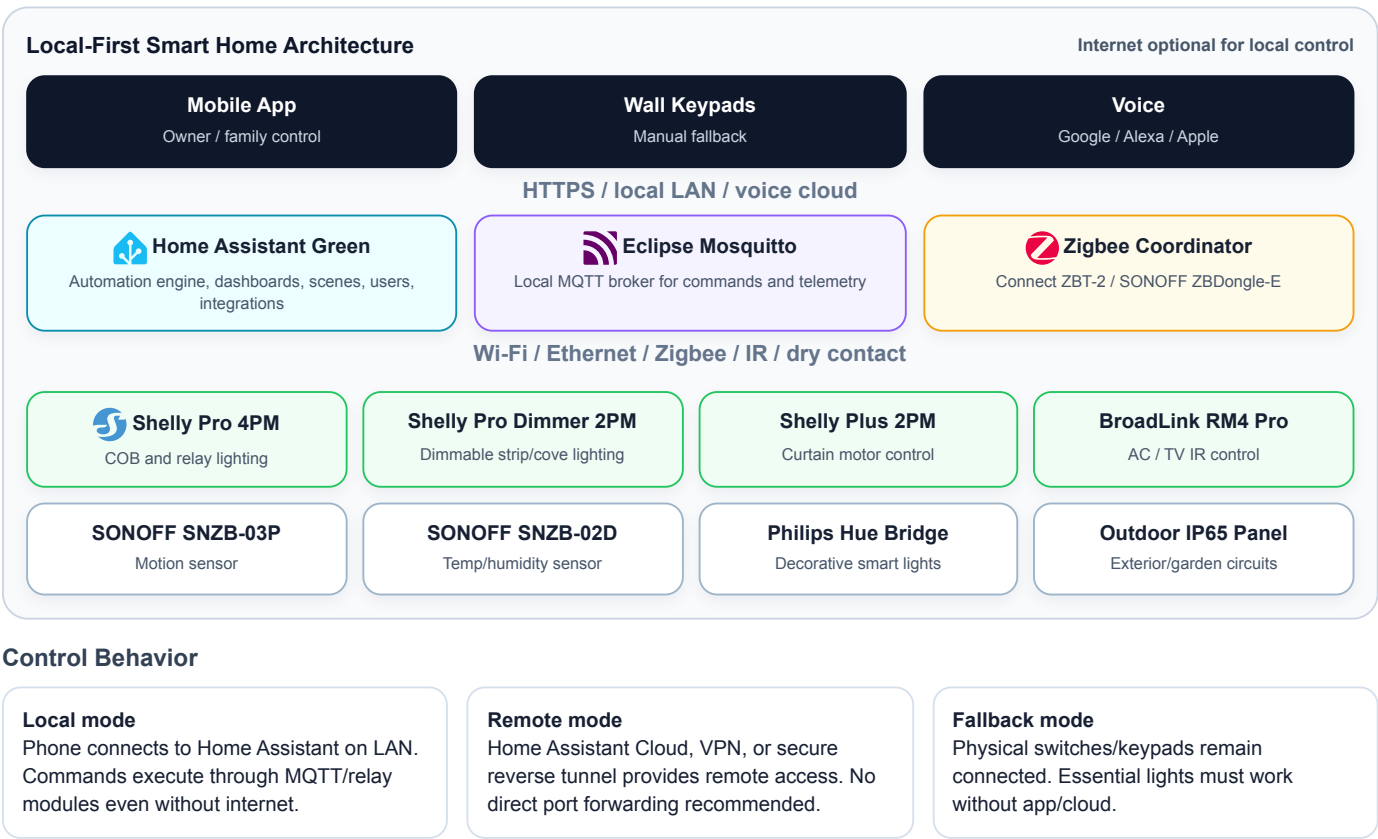
Control Hardware		
Function	Recommended Device / Product	Use
DIN relay	Shelly Pro 4PM	Lighting zones, metered relay outputs
DIN dimmer	Shelly Pro Dimmer 2PM	Dimmable lighting circuits
Curtains	Shelly Plus 2PM or Aqara Roller Shade Driver E1	Motorized curtains / roller shades
AC / TV IR	BroadLink RM4 Pro	IR/RF control for AC, TV, AV

Network And Voice Layer

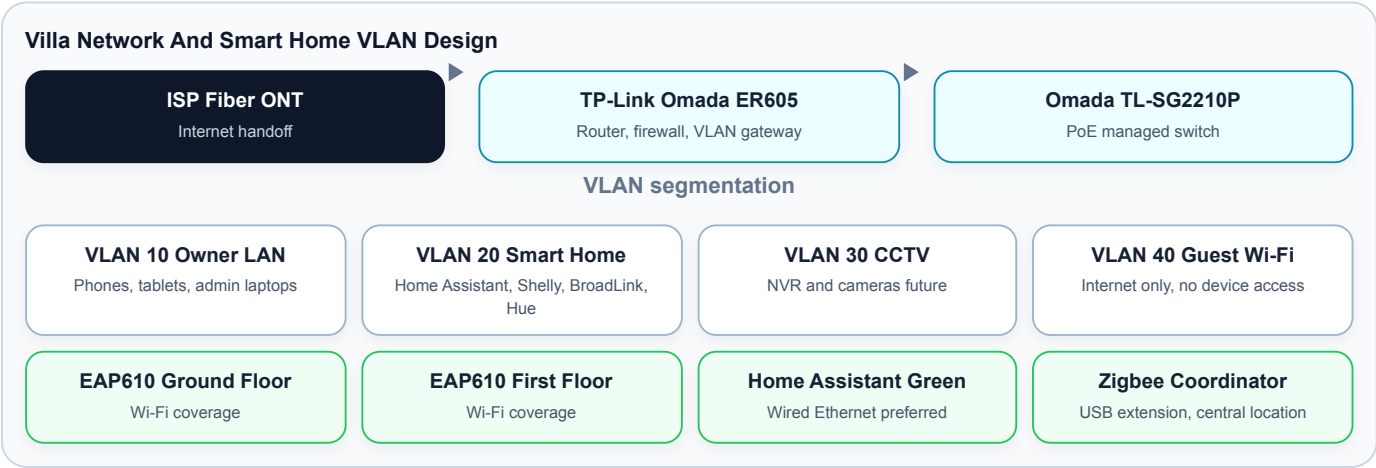
Function	Recommended Device / Product	Notes
Router / gateway	TP-Link Omada ER605	VLAN-ready router for smart home segmentation
PoE switch	TP-Link Omada TL-SG2210P	Gateway, access points, cameras, panels
Wi-Fi AP	TP-Link Omada EAP610 / EAP650	Reliable villa coverage
Voice	Google Home, Amazon Alexa, Apple Home	Via Home Assistant integrations/cloud bridge
Optional lighting ecosystem	Philips Hue Bridge / Hue Bridge Pro	Decorative lamps, strips, mood scenes

Electrical note: final relay/dimmer selection must follow load calculation, local electrical code, neutral availability, switchboard depth, heat dissipation, and DB/MCB design.

2. Architecture Diagram With Product Logos



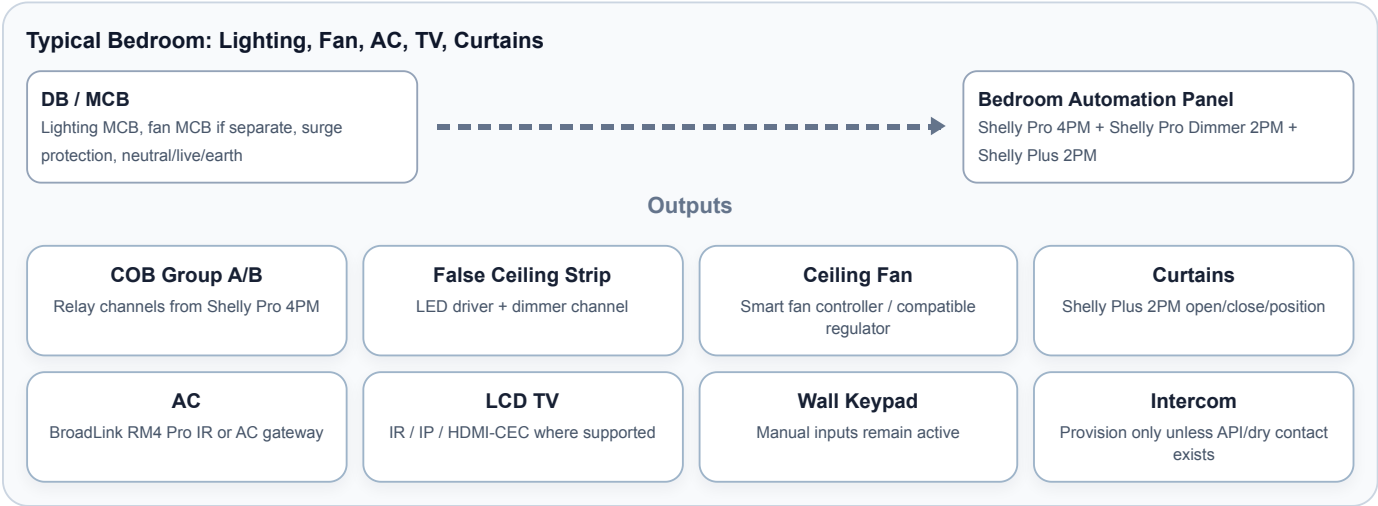
3. Network Topology Schematic



Network Rules

Rule	Reason
Home Assistant must be wired by Ethernet.	Stable command execution and reliable dashboards.
Smart devices stay on Smart Home VLAN.	Limits exposure and keeps guest devices isolated.
No router port-forwarding to Home Assistant.	Use Home Assistant Cloud, VPN, or secure tunnel instead.
Zigbee coordinator should use USB extension cable.	Reduces RF interference from gateway/USB port.

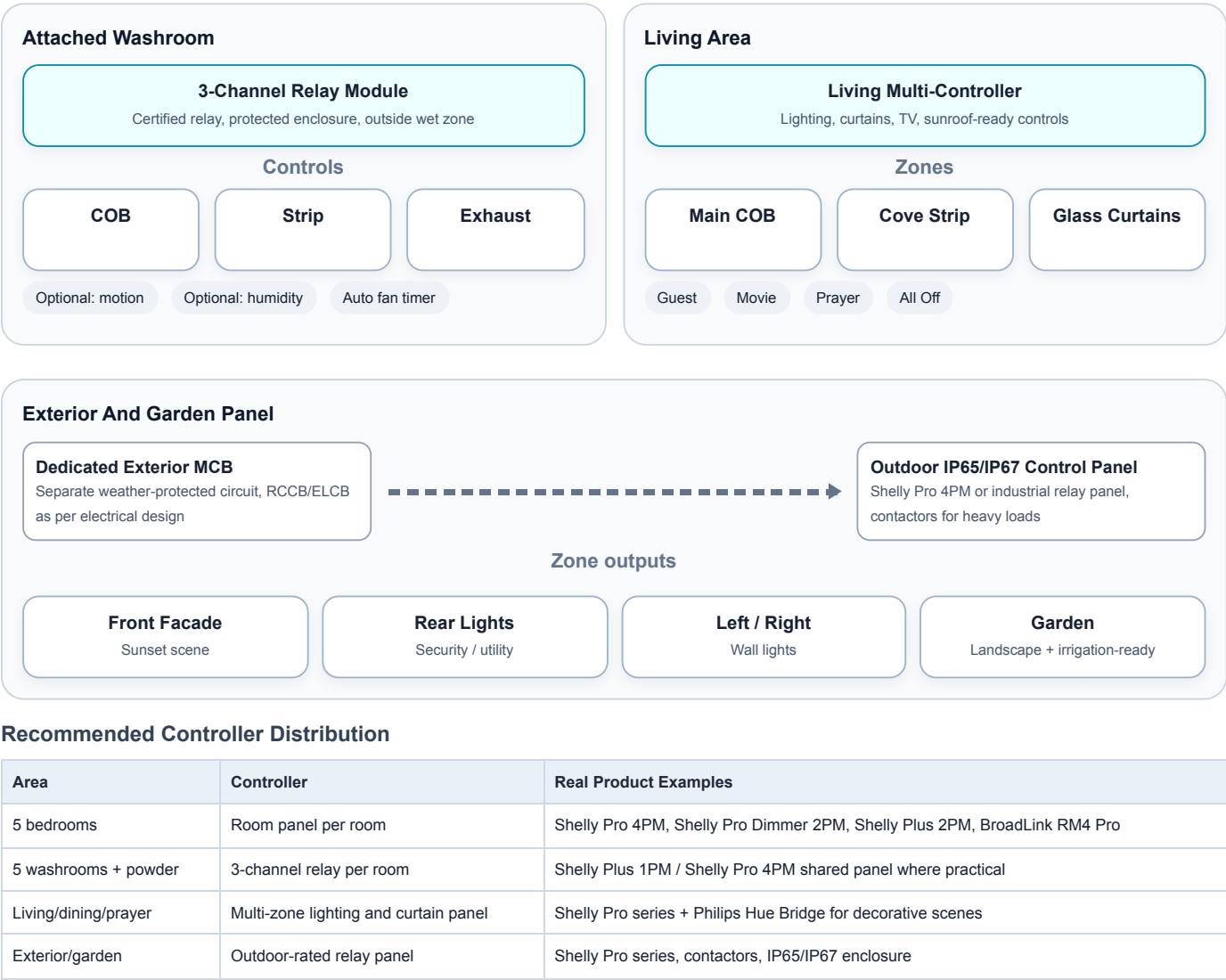
4. Bedroom Control Schematic



Bedroom Scene Logic

Scene	Lighting	Fan / AC	Curtain / TV
Relax	Strip warm 35%, COB low	Fan medium, AC comfort	Curtains partial close
Sleep	All lights off except optional night strip	Fan/AC sleep mode	Curtains close, TV off
Movie	COB off, strip dim	Comfort mode	Curtains close, TV on
Wake Up	Soft strip on	AC adjust	Curtains open

5. Washroom, Living, Exterior Schematics



6. Software, Naming, MQTT And Security

Room Naming Standard

Format: floor / area / device / function

- first/bedroom1/light/cob
- first/bedroom1/light/strip
- first/bedroom1/fan/main
- first/bedroom1/ac/main
- first/washroom1/exhaust/main
- exterior/front/light/facade

MQTT Topic Pattern

- home/{floor}/{area}/{device}/{function}/set
- home/{floor}/{area}/{device}/{function}/state
- home/{floor}/{area}/{device}/{function}/telemetry
- home/{floor}/{area}/scene/{scene}/trigger

Local Command Flow

Mobile App

Home Assistant

MQTT

Room Controller

State feedback returns to dashboard after execution

Security Requirements

Control	Requirement
Remote access	Home Assistant Cloud, VPN, or secure tunnel. No public port forwarding.
User roles	Owner/admin, family, guest, maintenance access.
Network	Smart Home VLAN isolated from Guest Wi-Fi and CCTV VLAN.
Backups	Scheduled Home Assistant backup to local NAS/cloud storage.
Manual fallback	All critical lights must remain physically operable.

Implementation Phases

Phase	Deliverable
1. Site survey	Device count, switchboard depth, DB mapping, curtain motor count, AC brands, Wi-Fi heatmap.
2. Electrical design	Load schedule, relay/dimmer map, MCB/RCCB plan, enclosure placement.
3. Network setup	Router, switch, access points, VLANs, gateway location, Zigbee coordinator placement.
4. Installation	Modules, sensors, curtain motors, IR gateways, panels, labels.
5. Commissioning	Room testing, offline test, remote test, scene test, handover document.

7. BOQ Categories And Final Recommendation

Category	Recommended Product Families	Notes
Automation gateway	Home Assistant Green / Intel NUC + Home Assistant OS	Use wired Ethernet and UPS.
Zigbee coordinator	Home Assistant Connect ZBT-2 / SONOFF ZBDongle-E	Use USB extension cable.
Relay control	Shelly Pro 4PM, Shelly Plus 1PM	Lighting and exhaust circuits after load check.
Dimming	Shelly Pro Dimmer 2PM, compatible LED drivers	Confirm dimmable driver type.
Curtains	Shelly Plus 2PM, Aqara Roller Shade Driver E1	Choose based on motor type and curtain hardware.
IR control	BroadLink RM4 Pro	AC and TV where no IP integration exists.
Sensors	SONOFF SNZB-03P, SNZB-02D, Aqara sensors	Motion, humidity, temperature, contact.
Network	TP-Link Omada ER605, TL-SG2210P, EAP610/EAP650	VLAN-ready and scalable.
Lighting ecosystem	Philips Hue Bridge / Hue Bridge Pro	Optional for decorative/mood lighting.

Final Architecture Recommendation

Build the solution around a local Home Assistant gateway with MQTT, VLAN-separated networking, room-wise smart relay/dimmer panels, Zigbee sensors, and secure remote access. Keep all daily functions local: lights, fans, curtains, AC scenes, washroom exhaust, staircase lights, and exterior/garden scenes.

The strongest design is not a cloud-only smart home. It is a professionally wired local automation system with app, voice, scenes, and remote access layered on top.

References Used For Product Naming

Product names and capabilities were checked against official product pages where available: Home Assistant Green, SONOFF ZBDongle-E, Aqara Roller Shade Driver E1, Philips Hue Bridge, and TP-Link Omada ER605.